

DCCN Lab Assignment – 3

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Answer the following questions for captured file dns1.pcap (DNS Protocol)

1. Locate the DNS query and response messages. Are they sent over UDP or TCP?

8	2004-08-31 03:27:43.582042	128.238.38.160	128.238.29.23	DNS	72 Standard query 0x006e A www.ietf.org
9	2004-08-31 03:27:43.582886	128.238.29.23	128.238.38.160	DNS	104 Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

DNS Query

8 2004-08-31 03:27:43.582042 128.238.38.160 128.238.29.23 DNS 72
Standard query 0x006e A www.ietf.org

DNS Query Response

9 2004-08-31 03:27:43.582886 128.238.29.23 128.238.38.160 DNS 104
Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

They are sent over UDP (User Datagram Protocol).

2. What is the destination port for the DNS query message? What is the source port of DNS response message?

Destination Port of DNS Query Message: 53

Source Port of DNS Response Message: 53

3. To what IP address is the DNS query message sent? Use nm-tool command to determine the IP address of your local DNS server. Are these two IP addresses the same?

DNS Query message is sent to 128.238.29.23

IP address of my local DNS server is 192.168.1.250 and 192.168.1.249

No, these addresses are not the same.

```
sid ~  
→ nmcli | grep -A 2 "DNS"  
DNS configuration:  
    servers: 192.168.1.250 192.168.1.249  
    interface: wlp0s20f3
```

4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

This is a type “A” query. This type of query is used to retrieve addresses for IPv4. No the Answer RRs field is 0 so the query doesn’t contain any “answers”.

```
‣ Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)  
‣ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-MSRP-rou  
‣ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23  
‣ User Datagram Protocol, Src Port: 3163, Dst Port: 53  
‣ Domain Name System (query)  
    Transaction ID: 0x006e  
    ‣ Flags: 0x0100 Standard query  
      Questions: 1  
      Answer RRs: 0  
      Authority RRs: 0  
      Additional RRs: 0  
    ‣ Queries  
      ‣ www.ietf.org: type A, class IN  
        Name: www.ietf.org  
        [Name Length: 12]  
        [Label Count: 3]  
        Type: A (1) (Host Address)  
        Class: IN (0x0001)  
        [Response In: 9]
```

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

2 answers are provided in the DNS response message.

The DNS server is returning two valid IPv4 addresses for www.ietf.org. This is common and is typically used for load balancing, redundancy, or geographic distribution.

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message?

Destination Address of SYN Packet : 132.151.6.75

Yes the destination IP address of the SYN packet corresponds to the first IP address provided in the DNS response message

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

No, it uses HTTP queries to get images

```
31 2004-08-31 03:27:43.699066 128.238.38.160 132.151.6.75 HTTP 314
GET /images/blue.gif HTTP/1.1
```

Use the command: nslookup -type=NS mit.edu (Use dns2.pcap file)

8. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

The DNS query message is being sent to 128.238.29.22

No, IP address of my local DNS server is 192.168.1.250 and 192.168.1.249

9. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

There are 3 DNS queries. Details are listed below:

The first DNS query is of type “PTR”. No, it does not contain any “answers”.

```

> Frame 488: 86 bytes on wire (688 bits), 86 bytes captured
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst:
> Internet Protocol Version 4, Src: 128.238.38.160, Dst:
> User Datagram Protocol, Src Port: 3744, Dst Port: 53
> Domain Name System (query)
  Transaction ID: 0x0001
  Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  Queries
    - 22.29.238.128.in-addr.arpa: type PTR, class IN
      Name: 22.29.238.128.in-addr.arpa
      [Name Length: 26]
      [Label Count: 6]
      Type: PTR (12) (domain name PoinTeR)
      Class: IN (0x0001)
      [Response In: 489]
```

The second DNS query is of type “NS”. No, it does not contain any “answers”.

```

> Frame 490: 76 bytes on wire (608 bits), 76 bytes captured
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst:
> Internet Protocol Version 4, Src: 128.238.38.160, Dst:
> User Datagram Protocol, Src Port: 3745, Dst Port: 53
> Domain Name System (query)
  Transaction ID: 0x0002
  Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  Queries
    - mit.edu.poly.edu: type NS, class IN
      Name: mit.edu.poly.edu
      [Name Length: 16]
      [Label Count: 4]
      Type: NS (2) (authoritative Name Server)
      Class: IN (0x0001)
      [Response In: 491]
```

The third DNS query is of type "NS". No, it does not contain any "answers".

```

> Frame 492: 67 bytes on wire (536 bits), 67 bytes captured
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst:
> Internet Protocol Version 4, Src: 128.238.38.160, Dst:
> User Datagram Protocol, Src Port: 3746, Dst Port: 53
- Domain Name System (query)
  Transaction ID: 0x0003
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  - Queries
    - mit.edu: type NS, class IN
      Name: mit.edu
      [Name Length: 7]
      [Label Count: 2]
      Type: NS (2) (authoritative Name Server)
      Class: IN (0x0001)
      [Response In: 493]
```

10. Examine the DNS response message. What MIT nameservers does the response message provide? Does this response message also provide the IP addresses of the MIT nameservers?

The first DNS response doesn't provide any MIT nameserver.

```

> Frame 489: 118 bytes on wire (944 bits), 118 bytes captured (944 bits)
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99
> Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160
> User Datagram Protocol, Src Port: 53, Dst Port: 3744
- Domain Name System (response)
  Transaction ID: 0x0001
  > Flags: 0x8580 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 0
  Additional RRs: 0
  > Queries
  - Answers
    - 22.29.238.128.in-addr.arpa: type PTR, class IN, dns-prime.poly.edu
      Name: 22.29.238.128.in-addr.arpa
      Type: PTR (12) (domain name Pointer)
      Class: IN (0x0001)
      Time to live: 3600 (1 hour)
      Data length: 20
      Domain Name: dns-prime.poly.edu
      [Request In: 488]
      [Time: 0.000367000 seconds]
```

The second DNS response provides the poly.edu nameserver. IP address is not provided

```

> Frame 491: 135 bytes on wire (1080 bits), 135 bytes captured
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IB
> Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238
> User Datagram Protocol, Src Port: 53, Dst Port: 3745
- Domain Name System (response)
  Transaction ID: 0x0002
  Flags: 0x8583 Standard query response, No such name
  Questions: 1
  Answer RRs: 0
  Authority RRs: 1
  Additional RRs: 0
  Queries
  - Authoritative nameservers
    - poly.edu: type SOA, class IN, mname dns-prime.poly.edu
      Name: poly.edu
      Type: SOA (6) (Start Of a zone of Authority)
      Class: IN (0x0001)
      Time to live: 3600 (1 hour)
      Data length: 39
      Primary name server: dns-prime.poly.edu
      Responsible authority's mailbox: admin
      Serial Number: 204503
      Refresh Interval: 1800 (30 minutes)
      Retry Interval: 900 (15 minutes)
      Expire limit: 604800 (7 days)
      Minimum TTL: 900 (15 minutes)
[Request In: 490]
[Time: 0.000344000 seconds]
```

The third DNS response returns the IP addresses of 3 authoritative name servers.

```

  Authority RRs: 0
  Additional RRs: 3
  Queries
  - Answers
    - mit.edu: type NS, class IN, ns bitsy.mit.edu
      Name: mit.edu
      Type: NS (2) (authoritative Name Server)
      Class: IN (0x0001)
      Time to live: 20736 (5 hours, 45 minutes, 36 seconds)
      Data length: 8
      Name Server: bitsy.mit.edu
    - mit.edu: type NS, class IN, ns strawb.mit.edu
      Name: mit.edu
      Type: NS (2) (authoritative Name Server)
      Class: IN (0x0001)
      Time to live: 20736 (5 hours, 45 minutes, 36 seconds)
      Data length: 9
      Name Server: strawb.mit.edu
    - mit.edu: type NS, class IN, ns w20ns.mit.edu
      Name: mit.edu
      Type: NS (2) (authoritative Name Server)
      Class: IN (0x0001)
      Time to live: 20736 (5 hours, 45 minutes, 36 seconds)
      Data length: 8
      Name Server: w20ns.mit.edu
  - Additional records
    - bitsy.mit.edu: type A, class IN, addr 18.72.0.3
      Name: bitsy.mit.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 20736 (5 hours, 45 minutes, 36 seconds)
      Data length: 4
      Address: 18.72.0.3
    - strawb.mit.edu: type A, class IN, addr 18.71.0.151
      Name: strawb.mit.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 20736 (5 hours, 45 minutes, 36 seconds)
      Data length: 4
      Address: 18.71.0.151
    - w20ns.mit.edu: type A, class IN, addr 18.70.0.160
      Name: w20ns.mit.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 20736 (5 hours, 45 minutes, 36 seconds)
      Data length: 4
      Address: 18.70.0.160
[Request In: 492]
[Time: 0.000361000 seconds]
```